Rakamin Academy x id/x partners

Prediksi Risiko Kredit Menggunakan Machine Learning Optimasi Keputusan Persetujuan Pinjaman Berbasis Data

ID/X Partners - Data Scientist

Presented by:
Mohammad Tyas Subianto

PROFILE

Mohammad Tyas Subianto
Data Science Undergraduate | Aspiring Data Engineer

Data Science undergraduate with hands-on experience in data preprocessing, transformation, and structured data preparation through industry-based virtual internship programs. Experienced in integrating multi-source datasets, performing data cleaning and validation, and building analysis-ready datasets to support analytical workflows.

Surabaya, Indonesia

mtyassubianto@gmail.com
linkedin.com/in/m-tyas-subianto

About Company

id/x partners is an Indonesian consulting firm specializing in Data Analytics, Decisioning, and RegTech solutions since 2006. With 18+ years of experience, we have helped 75+ institutions, including top banks, multifinance companies, fintechs, and insurers, leverage advanced analytics and AI-based solutions to solve challenges in digital lending, risk management, financial crime prevention, and regulatory compliance. This proven track record is recognized through multiple industry awards, including CIO Advisor's Top 10 APAC Data Analytics Consulting (2019) and four consecutive SAS Partner Appreciation Awards (2022-2025)

Background Problem

Institusi keuangan menghadapi tantangan dalam proses persetujuan pinjaman:

- Persetujuan pinjaman yang terlalu banyak meningkatkan risiko gagal bayar
- Penolakan pinjaman yang berlebihan mengurangi peluang keuntungan
- Penilaian kredit manual bersifat subjektif dan tidak konsisten

Permasalahan Utama

Bagaimana menyetujui pinjaman sebanyak mungkin tanpa meningkatkan risiko kredit?

Tujuan Proyek

Proyek ini bertujuan untuk:

- Memprediksi risiko gagal bayar nasabah
- Menghitung probabilitas risiko kredit setiap peminjam
- Mendukung keputusan persetujuan pinjaman berbasis data
- Mensimulasikan kebijakan risiko kred

Rakamin Academy

Dataset yang Digunakan

Dataset: LendingClub Loan Dataset

Dataset berisi informasi mengenai:

- Kondisi finansial peminjam
- Riwayat kredit
- Karakteristik pinjaman
- Status pembayaran pinjaman

Target utama: Loan Status

Dataset Shape: (466285, 75)

	Unnamed: 0	id	member_id	loan_amnt	funded_amnt	funded_amnt_inv	term
0	0	1077501	1296599	5000	5000	4975.0	36 months
1	1	1077430	1314167	2500	2500	2500.0	60 months
2	2	1077175	1313524	2400	2400	2400.0	36 months
3	3	1076863	1277178	10000	10000	10000.0	36 months
4	4	1075358	1311748	3000	3000	3000.0	60 months

5 rows × 75 columns

Definisi Risiko Kredit

Dua pendekatan risiko digunakan:

1. Conservative Risk

Kategori gagal bayar:

- Charged Off
- Default

➡ Merepresentasikan kerugian nyata.

2. Early Risk Detection

Kategori risiko:

- Charged Off
- Default
- Late Payment

➡ Mendeteksi potensi risiko lebih awal.

Rakamin Academy

Metodologi penelitian

CRISP-DM Phase	Project Implementation
Business Understanding	Defining credit risk prediction objectives
Data Understanding	Dataset exploration and feature analysis
Data Preparation	Data cleaning, leakage handling, encoding, and preprocessing
Modeling	Algorithm benchmarking using Logistic Regression, LightGBM, and XGBoost
Evaluation	Performance comparison and threshold optimization
Deployment	Risk scoring workflow and lending decision simulation

Rakamin Academy

Data Preparation

Tahap Persiapan Data

Langkah preprocessing data:

- Penanganan missing values
- Penghapusan fitur tidak relevan
- Transformasi fitur
- Encoding variabel kategorik
- Pembagian data training dan testing

Pipeline preprocessing meliputi:

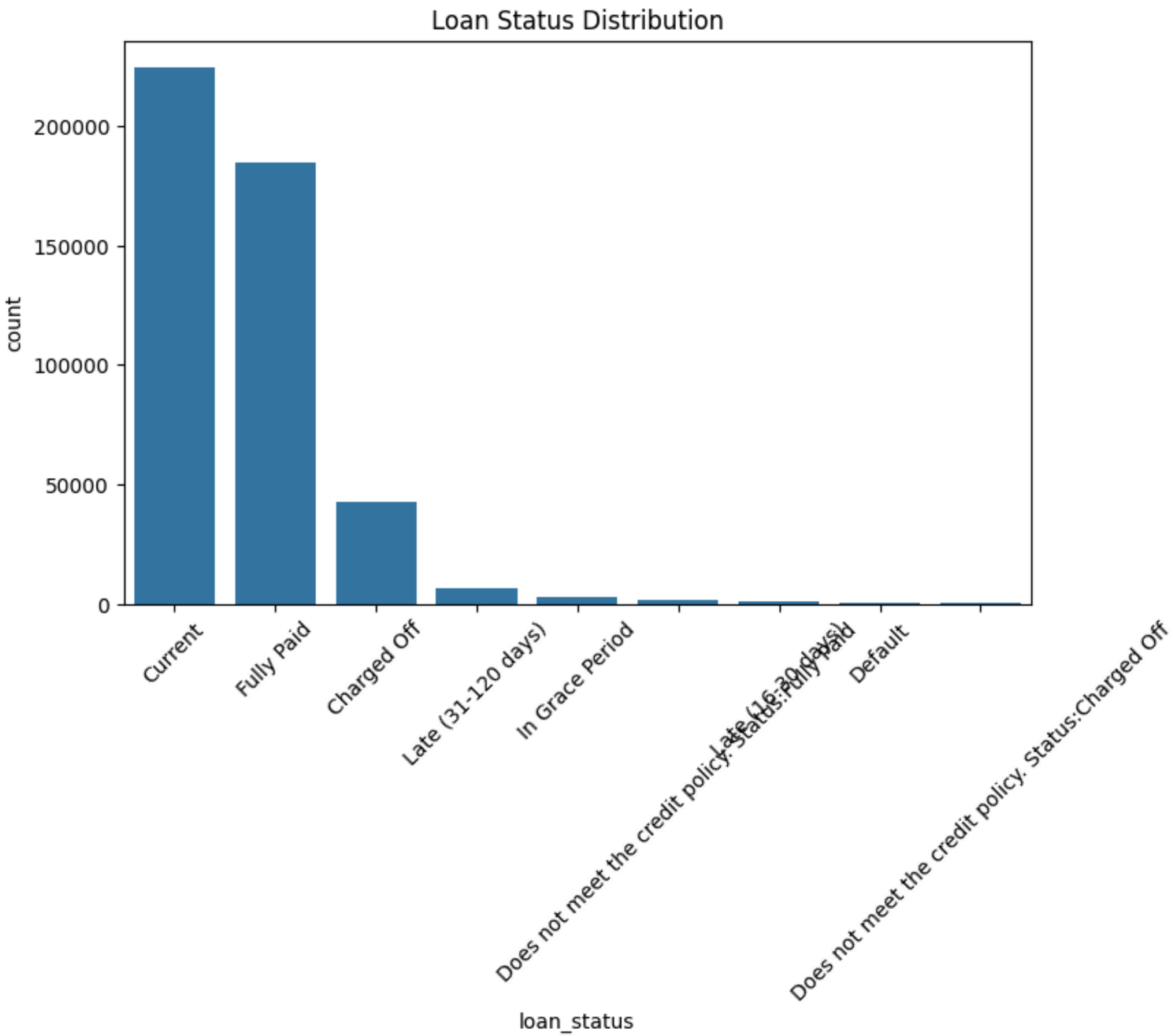
- Imputasi data
- Standardisasi
- One-Hot Encoding

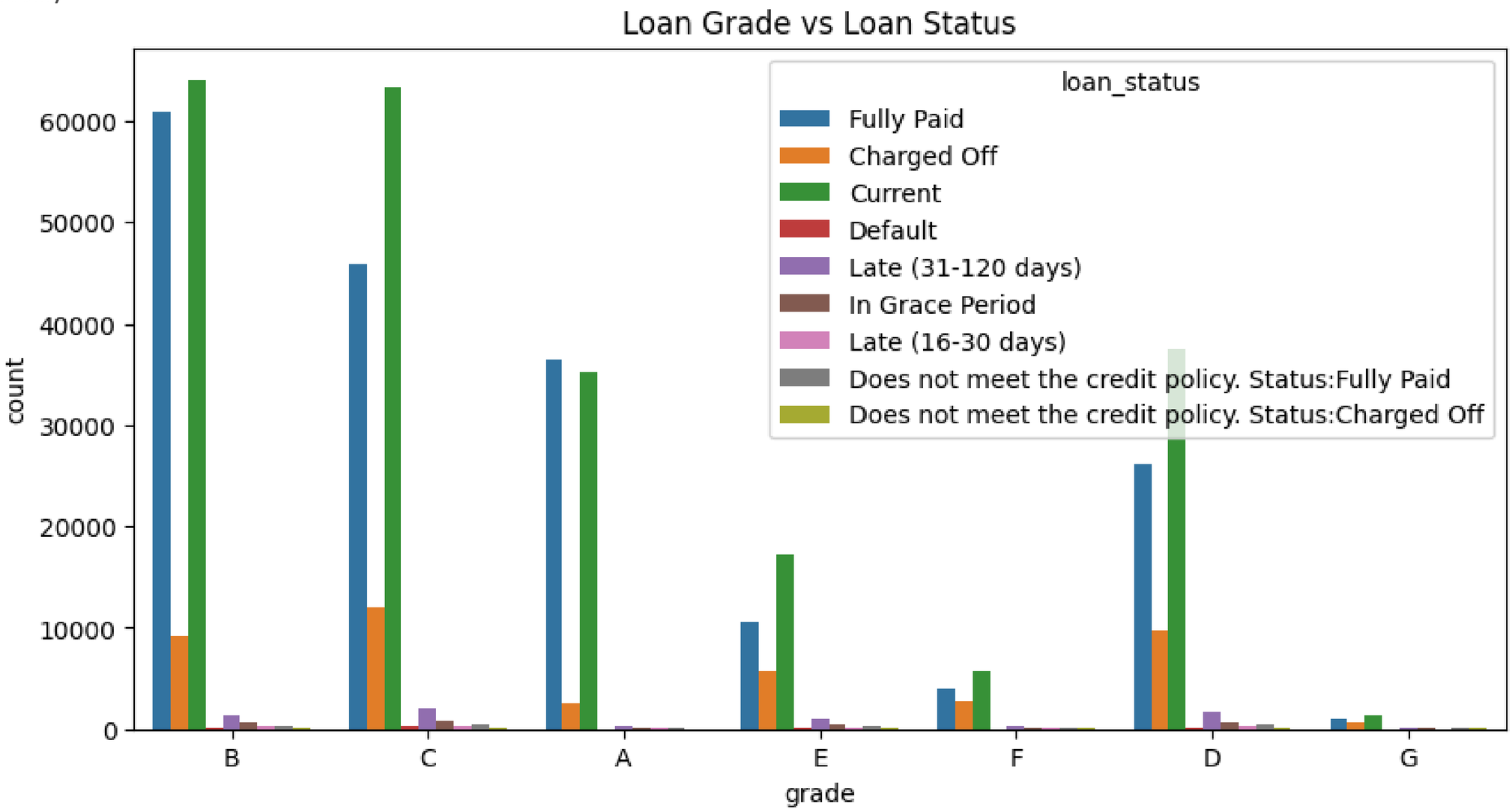
Rakamin Academy

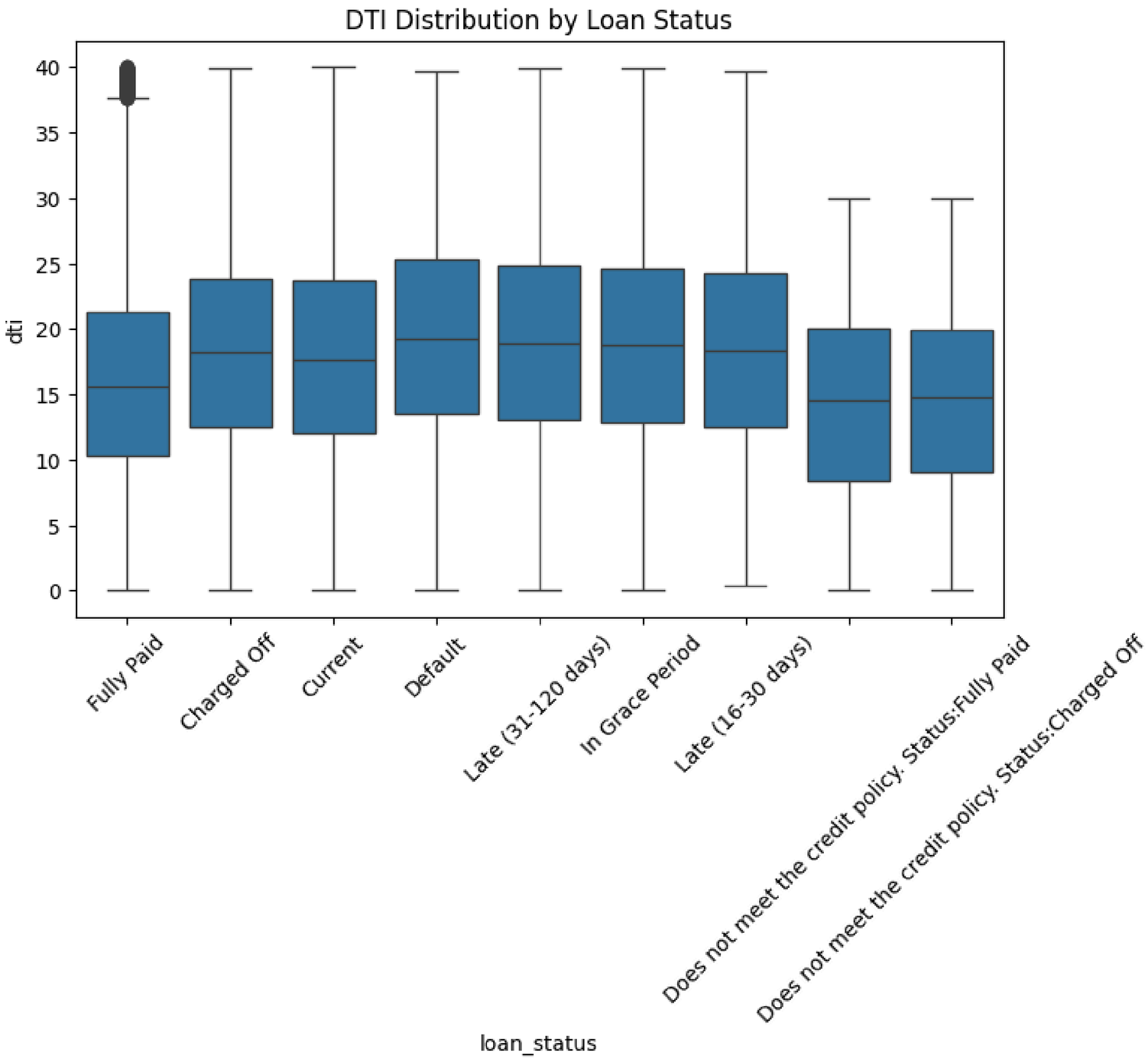
Exploratory Data Analysis (EDA)

EDA dilakukan untuk memahami pola risiko kredit.
Temuan utama:

- Suku bunga tinggi meningkatkan risiko gagal bayar
- Grade pinjaman rendah memiliki risiko lebih besar
- Debt-to-Income Ratio tinggi meningkatkan kemungkinan default


Loan Status Distribution
count
200000
150000
100000
50000
0
Current
Fully Paid
Charged Off
Late (31-120 days)
In Grace Period
Default
Does not meet the credit policy. Status:Charged Off
loan_status


Loan Grade vs Loan Status
loan_status
Fully Paid
Charged Off
Current
Default
Late (31-120 days)
In Grace Period
Late (16-30 days)
Does not meet the credit policy. Status:Fully Paid
Does not meet the credit policy. Status:Charged Off
count
0
10000
20000
30000
40000
50000
60000
B
C
A
E
F
D
G
grade


DTI Distribution by Loan Status
dti
40
35
30
25
20
15
10
5
0
Fully Paid
Charged Off
Current
Default
Late (31-120 days)
In Grace Period
Late (16-30 days)
Does not meet the credit policy. Status:Fully Paid
Does not meet the credit policy. Status:Charged Off
loan_status

Model Machine Learning

Model yang Digunakan

Tiga model dibandingkan:

- Logistic Regression
- XGBoost
- LightGBM

Tujuan:
Menentukan model terbaik dalam memprediksi risiko kredit.

Rakamin Academy

Perbandingan Performa Model

	mode	sample_n	model	bad_rate	roc_auc	pr_auc	precision_bad@0.5	recall_bad@0.5	f1_bad@0.5
1	early_risk	120000	LightGBM	0.21775	0.716516	0.407956	0.544554	0.105243	0.176395
2	early_risk	120000	XGBoost	0.21775	0.713866	0.405709	0.346448	0.650402	0.452085
0	early_risk	120000	LogReg	0.21775	0.707566	0.393968	0.381878	0.529851	0.443857

Model boosting memberikan performa terbaik dalam mendeteksi risiko kredit.

Rakamin Academy

Probabilitas Risiko

Pendekatan Berbasis Probabilitas

Model menghasilkan:

Probability of Default
Keputusan tidak langsung approve/reject,
tetapi berdasarkan tingkat risiko peminjam.

Simulasi Kebijakan Persetujuan Pinjaman

Kebijakan Agresif

- Approval Rate ≈ 68%
- Bad Rate ≈ 14%
- Risiko terdeteksi ≈ 56%

Kebijakan Konservatif

- Approval Rate ≈ 50%
- Bad Rate ≈ 11%
- Risiko terdeteksi ≈ 73%

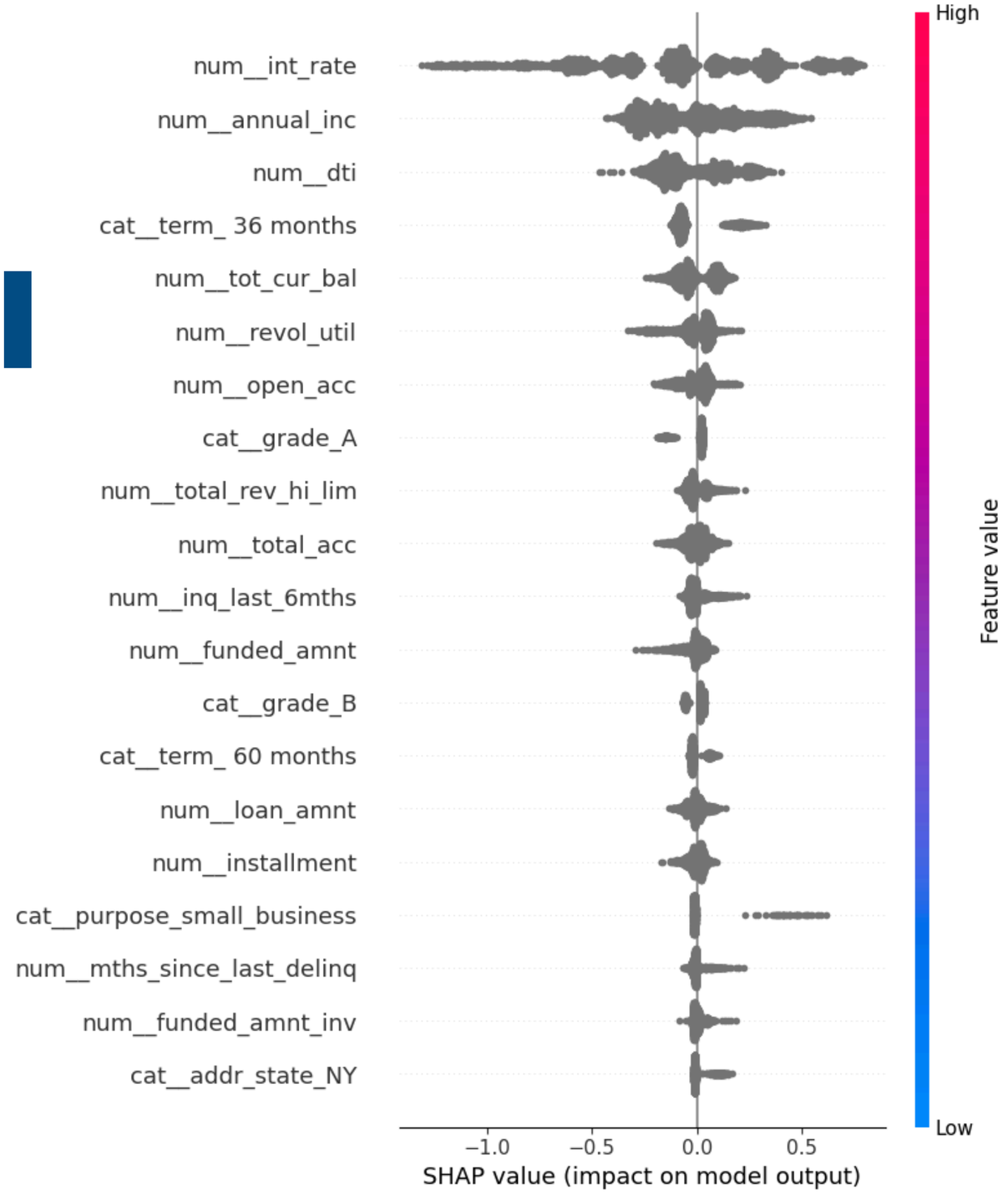
	policy	threshold	approval_rate	bad_rate_among_approved	bad_caught_rate
0	Risk Detection	0.55	0.679625	0.141745	0.557597
1	Conservative Approval	0.45	0.504125	0.113315	0.737658

Rakamin Academy

Explainable AI

Faktor Utama Risiko Kredit
Model mengidentifikasi faktor penting:

- Loan Grade
- Interest Rate
- Debt-to-Income Ratio
- Annual Income
- Loan Term



Rakamin Academy

Dampak Bisnis

Dampak Implementasi Model
Dengan Machine Learning:

- Evaluasi kredit lebih konsisten
- Risiko gagal bayar menurun
- Persetujuan pinjaman optimal
- Keputusan berbasis data

Kesimpulan

- Machine Learning mampu memprediksi risiko kredit secara efektif
- Model boosting memberikan performa terbaik
- Penyesuaian threshold memungkinkan strategi bisnis fleksibel
- Model mendukung keputusan persetujuan pinjaman yang lebih optimal

Insight Utama:

Prediksi risiko kredit tidak hanya meningkatkan akurasi, tetapi juga membantu pengambilan keputusan bisnis.

Rakamin Academy

Link Google Colab

https://colab.research.google.com/drive/1ETT87_qW229XvbMskE0f9IvLb7XBm3w-?usp=sharing

Rakamin Academy

Link Folder

https://drive.google.com/drive/folders/1xTfI63nw88G1eOKKcMLJO_mVCg6B-t6l?usp=drive_link

Link Github

https://github.com/Underag3/Credit_Risk_Prediction_End_to_End

Link Video Presentasi

https://drive.google.com/drive/folders/1BOXePCJQ1SFzgmq_qE_6FNYRA-xvnmYB?usp=drive_link